

Summer Camp: Drone Swarm Technologies

Systemarchitekturen & Steuerungskonzepte

22.09.2026

Prof. Dr. Simon Schwerd

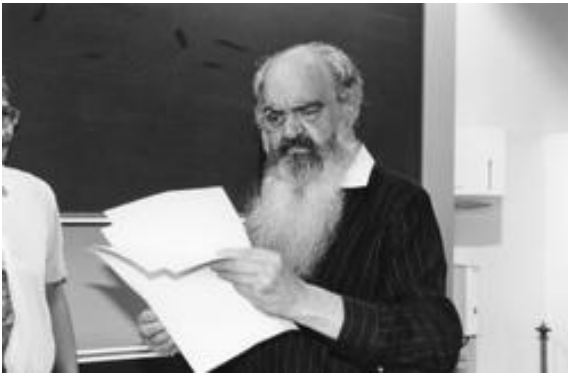
Inhalt

1. Architektur Gesamtsystem
2. Schlüsselkonzepte der UAV-Gesamtsystemarchitektur
3. Einführung in Multi-UAV & Swarming

Was bezeichnet Systemarchitektur?

“The word system stands for **connectivity**. By that we mean every accumulation of parts that stand in relation with each other.

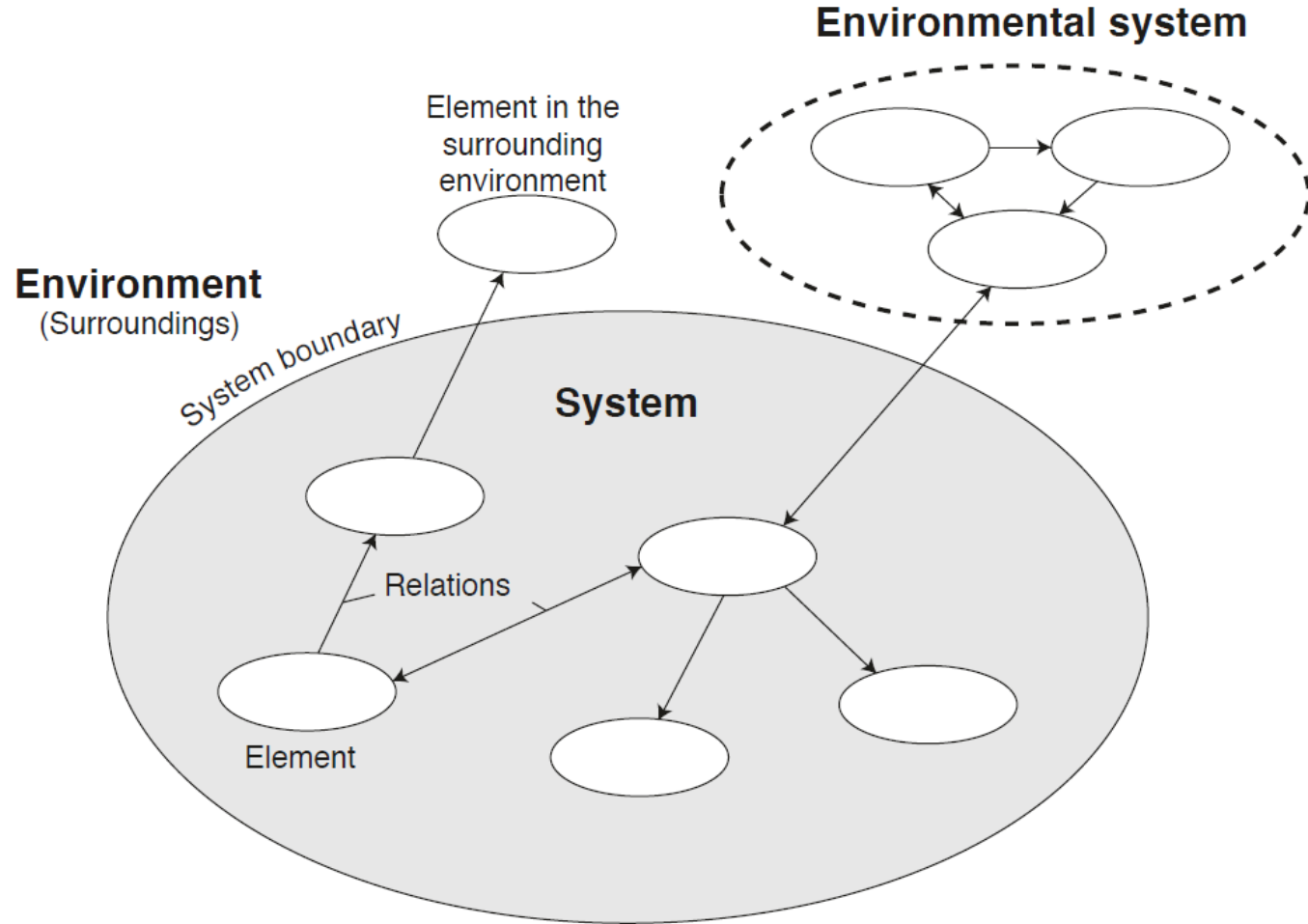
What we define as a system is therefore a system because it comprises parts that stand in relation with each other and in a certain respect **form a whole.**”



Stafford Beer

Was ist die Architektur eines Systems?

- Kernbegriffe:
 - Systeme & Sub-systeme
 - Systemgrenze
 - Relationen
 - Elemente mit
 - Eigenschaften
 - Funktionen
 - *Aspect* (Perspektive)



Haberfellner, Reinhard, et al. *Systems Engineering*. Cham: Springer International Publishing, 2019.

Sprache einer Systemarchitektur in der Softwareentwicklung (& darüber hinaus)

Werkzeuge:

- Boxen
- Pfeile
- Fachvokabular (z.B. Queues, Bus, Server...)

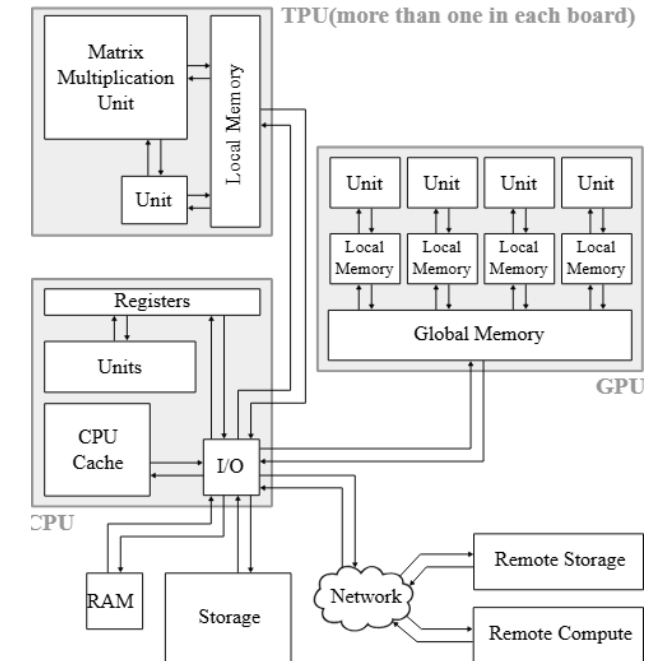
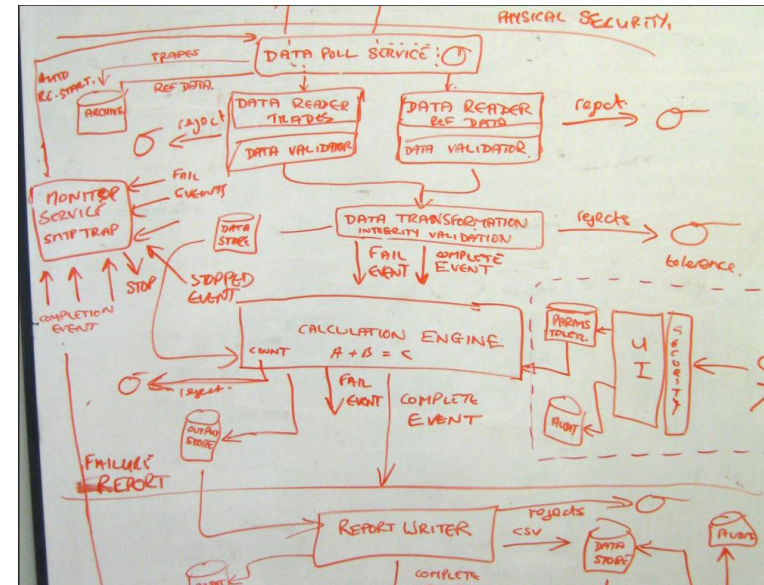
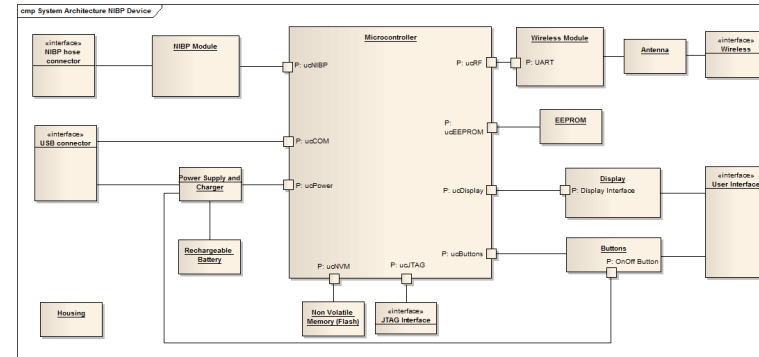
Modellierungssprachen:

- Unified Modeling Language, UML
- SysML
- ...

Technische Systemarchitekturen in der Mechatronik/Robotik

Aspekte

- Systemarchitektur
- Hardwarearchitektur
- Softwarearchitektur
- Regelungsarchitektur
- Kommunikationsarchitektur
- Funktionsarchitektur
- Integrationsarchitektur
- ...



Aufgabe

Gruppe 1:
Zeichnen Sie die
Systemarchitektur eines UAVs
auf!

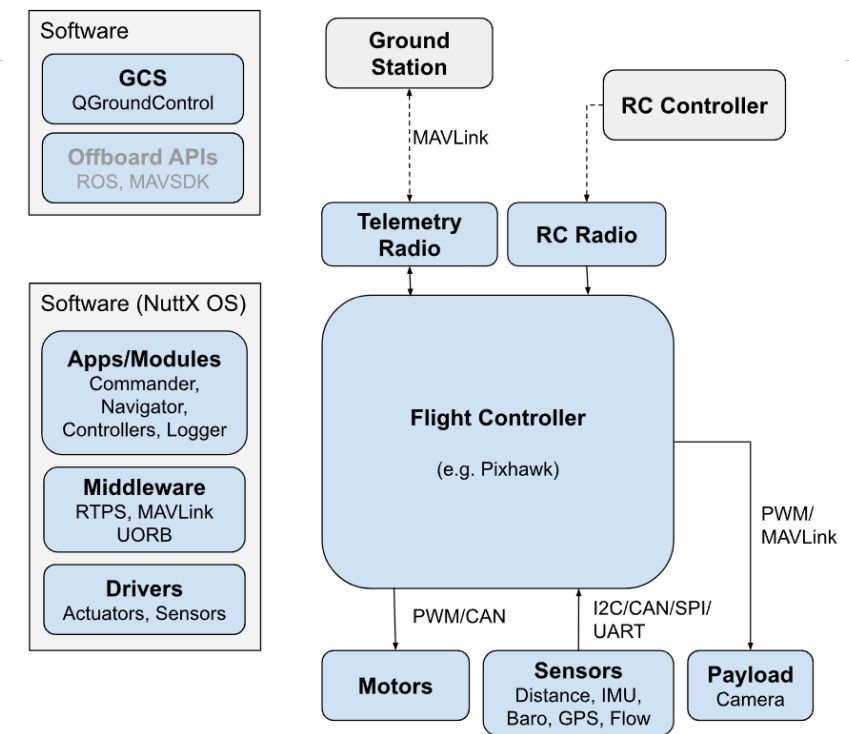
Gruppe 2:
Zeichnen Sie die
Hardwarearchitektur eines UAVs
auf!

Gruppe 3:
Zeichnen Sie die
Softwarearchitektur eines UAVs
auf!



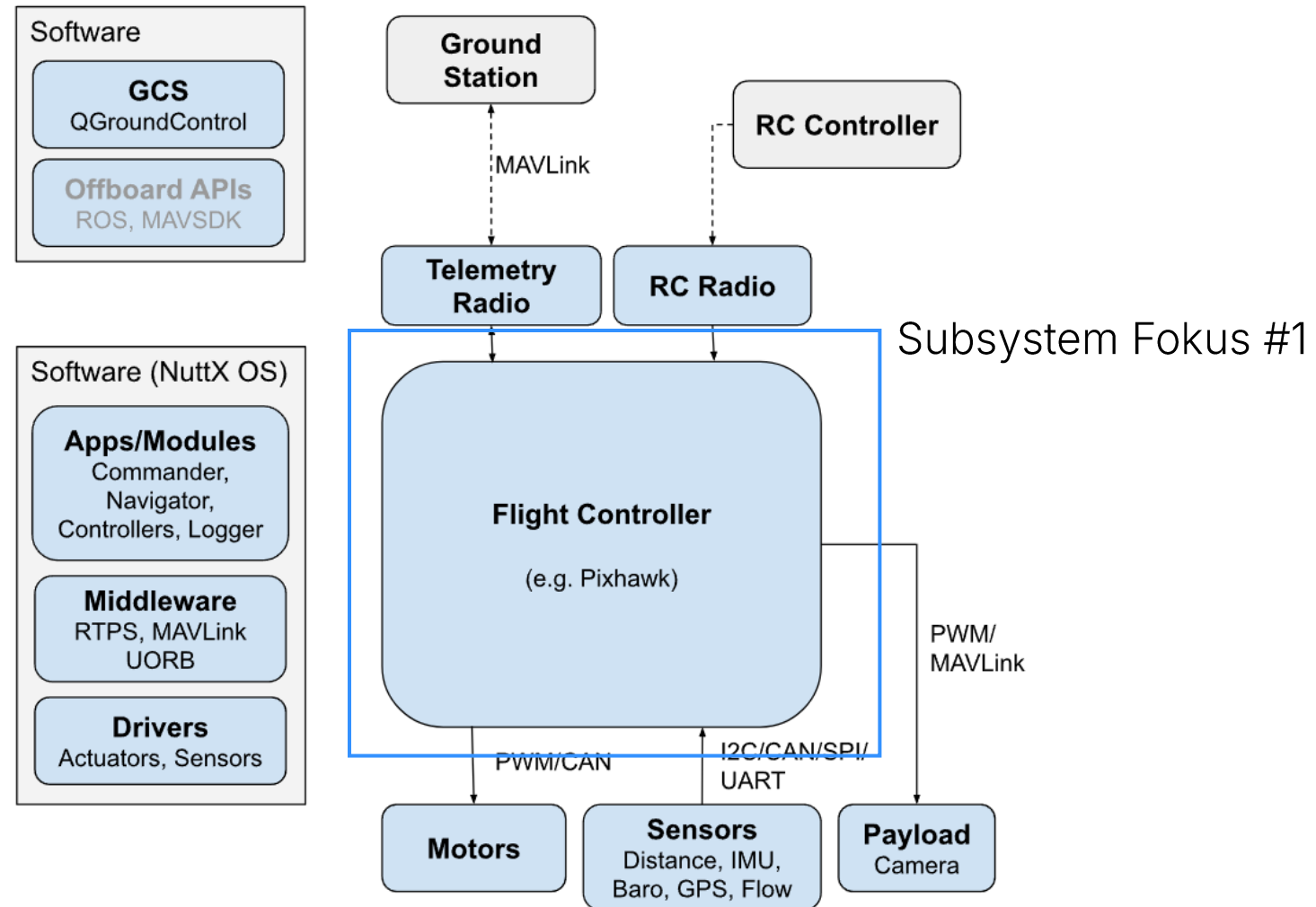
Case Study: PX4 System Architecture

- PX4 ist ein Autopilot für Drohnen und unbemannte Plattformen (Multicopter, VTOL, Helikopter, ...)
- PX4 unterstützt eine Vielzahl von Features:
 - manuelle und „autonome“ Flight Modes
 - QGroundControl als Bodenkontrollstation
 - Simulations- & Testingframeworks
- Ausführliche Dokumentation



<https://docs.px4.io/main/en/concept/architecture>

System Architecture



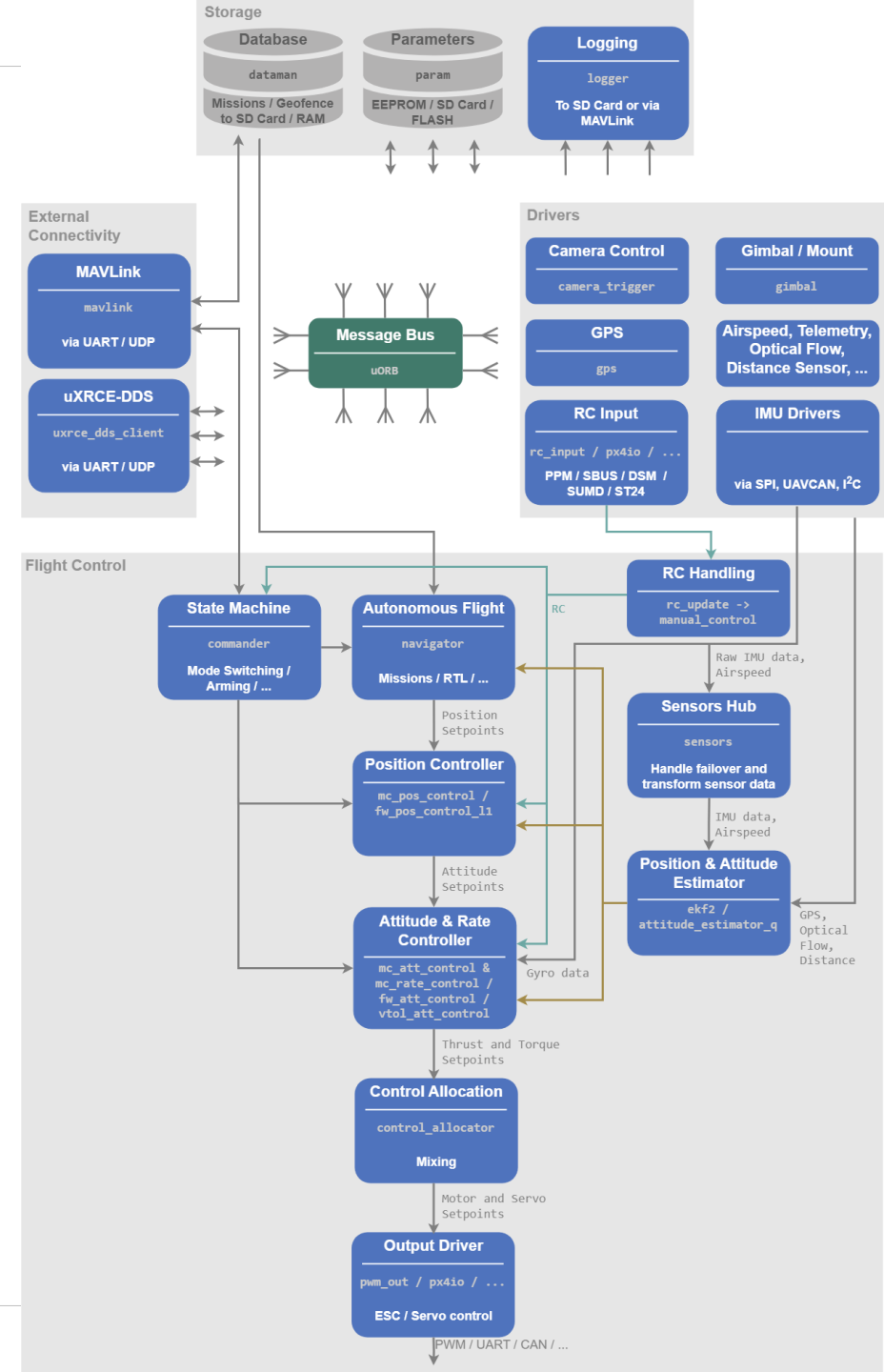
<https://docs.px4.io/main/en/concept/architecture>

Systemarchitektur: Software

Systemarchitektur am Beispiel von Px4

- Flight Control
- Middleware
- HW-Driver
- Message Bus
- Simulation Layer
- Storage
- External Connectivity

<https://docs.px4.io/main/en/concept/architecture>



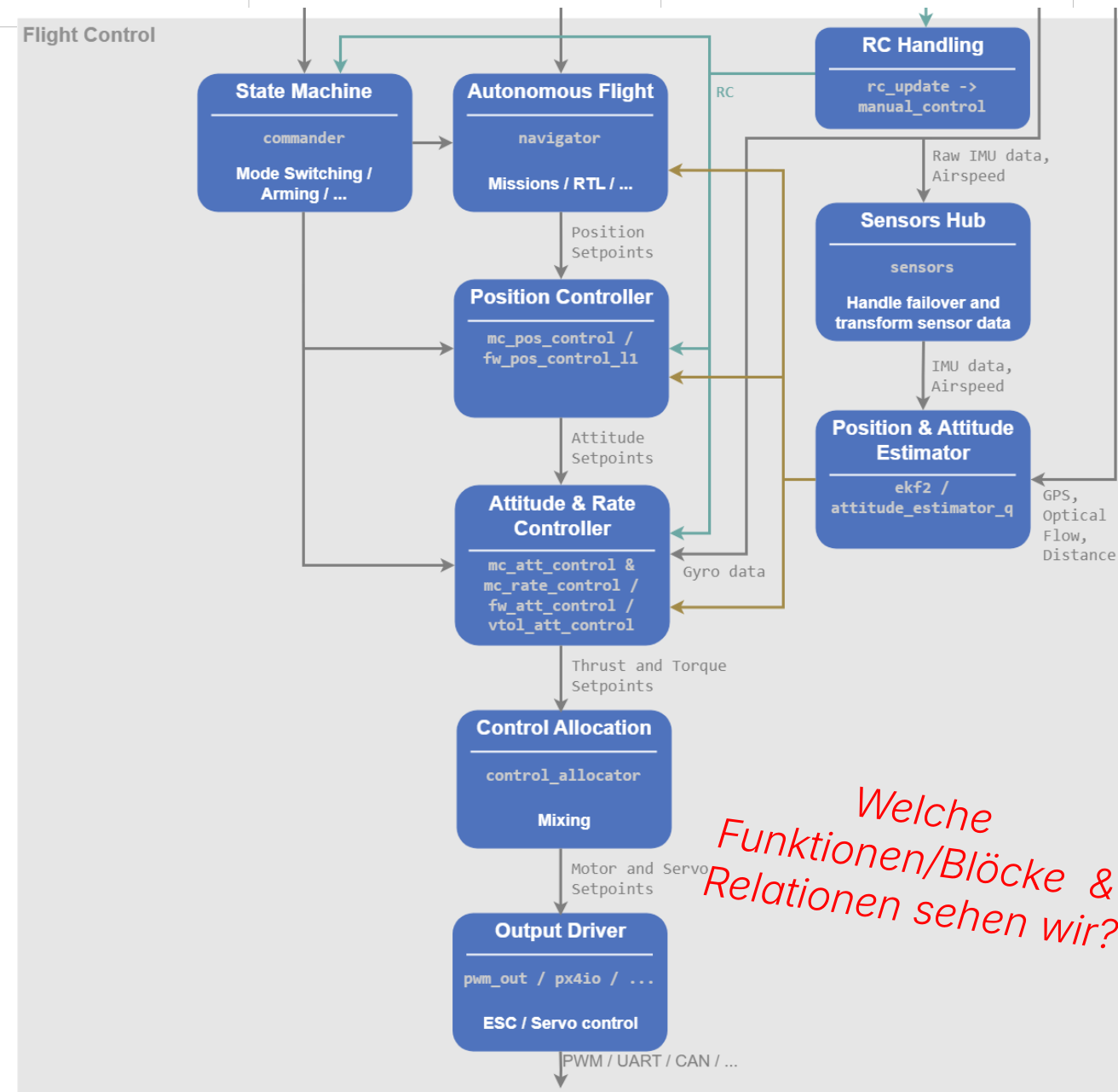
Systemarchitektur: Software

Systemarchitektur am Beispiel von Px4

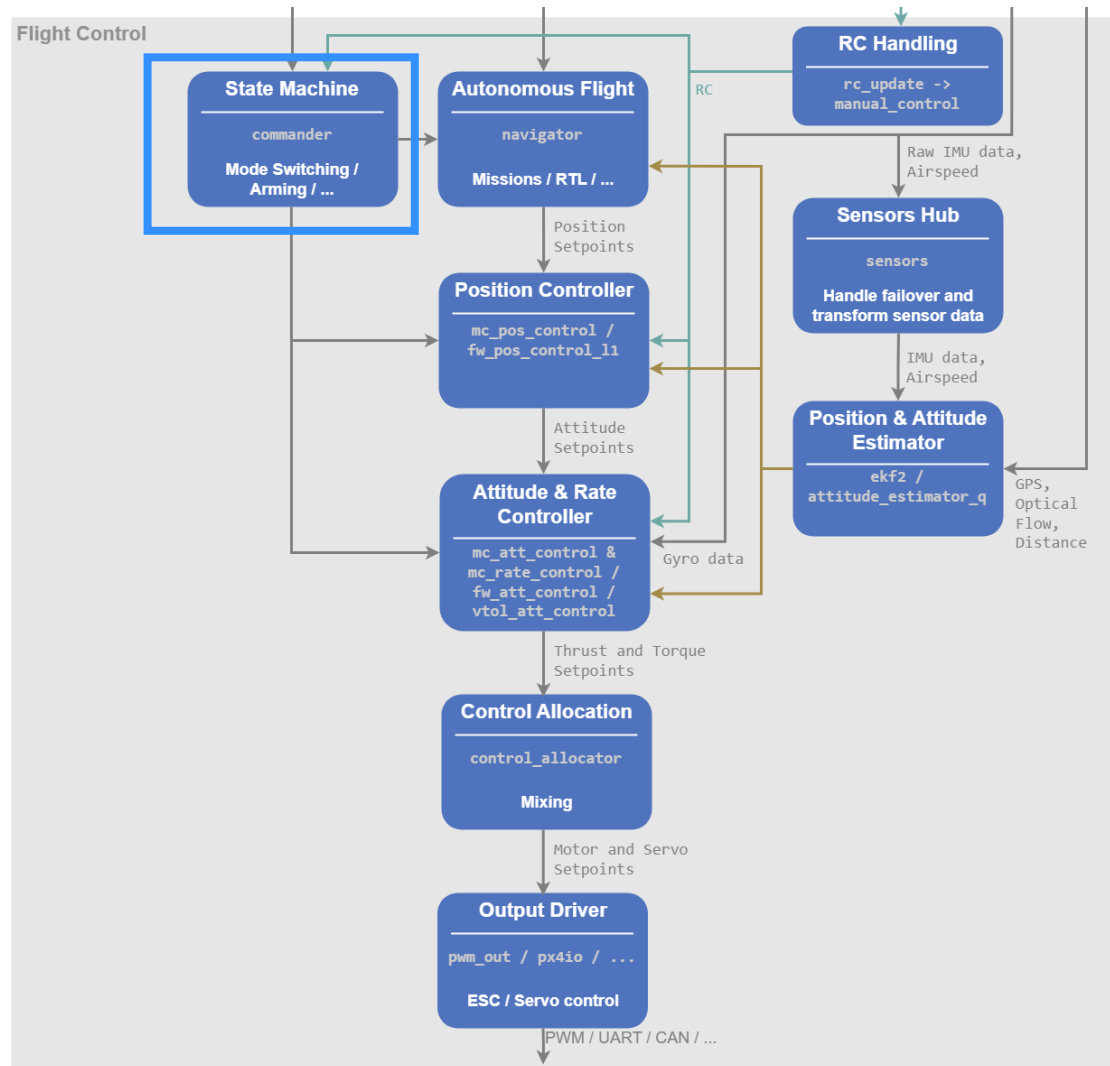
• Flight Control

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Regelungsarchitektur: Flight Stack

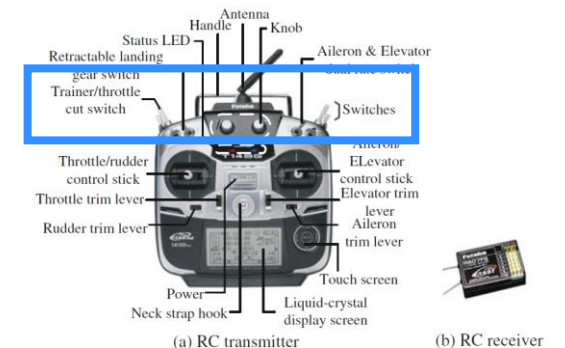


Funktion State Machine:

Festlegung, welche Eingangsgrößen der Flight Stack berücksichtigt (bspw. Wegpunkte vs. RC)

Kann manuell über RC gesteuert werden, kann aber auch über andere Softwarekomponenten (bspw. Companion Computer) getriggert werden.

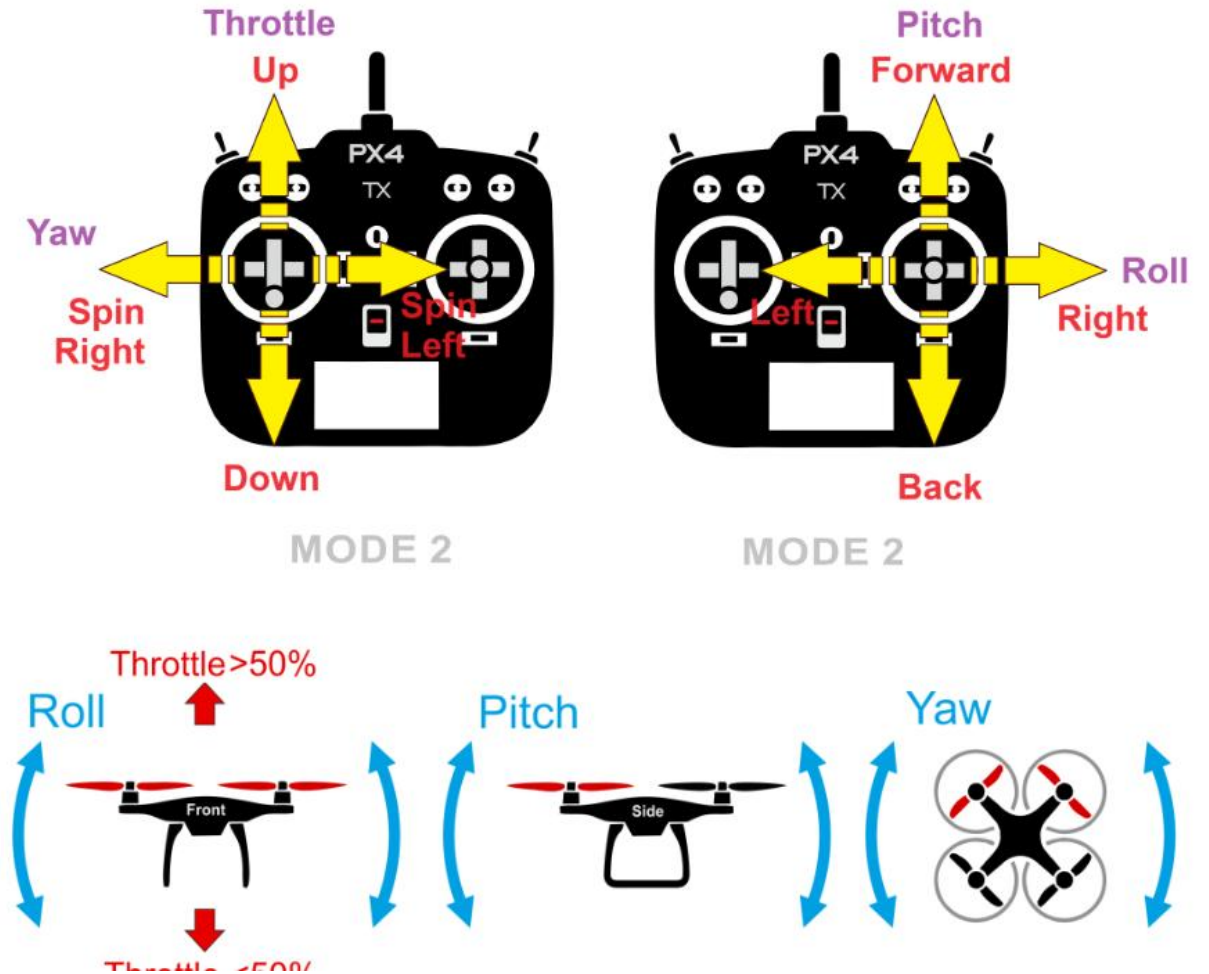
Fig.2.20 Futaba RC transmitter and receiver



<https://docs.px4.io/main/en/concept/architecture>

Modes & Arming

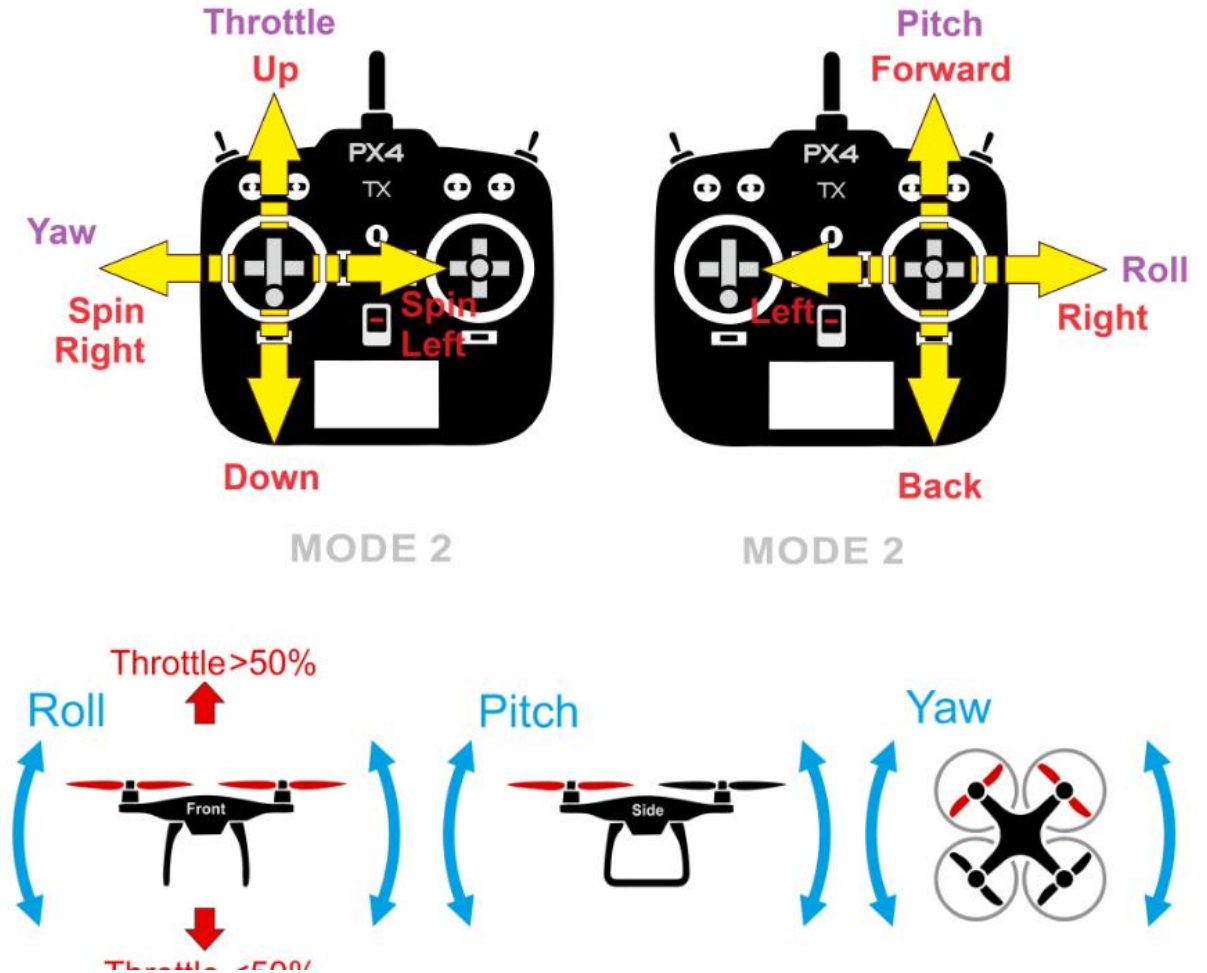
- Flight Modes können manuell oder autonom sein
- Manual
 - Position mode
 - Altitude mode
 - Stabilized mode
 - Acro



https://docs.px4.io/v1.16/en/flying/basic_flying_mc

Modes & Arming

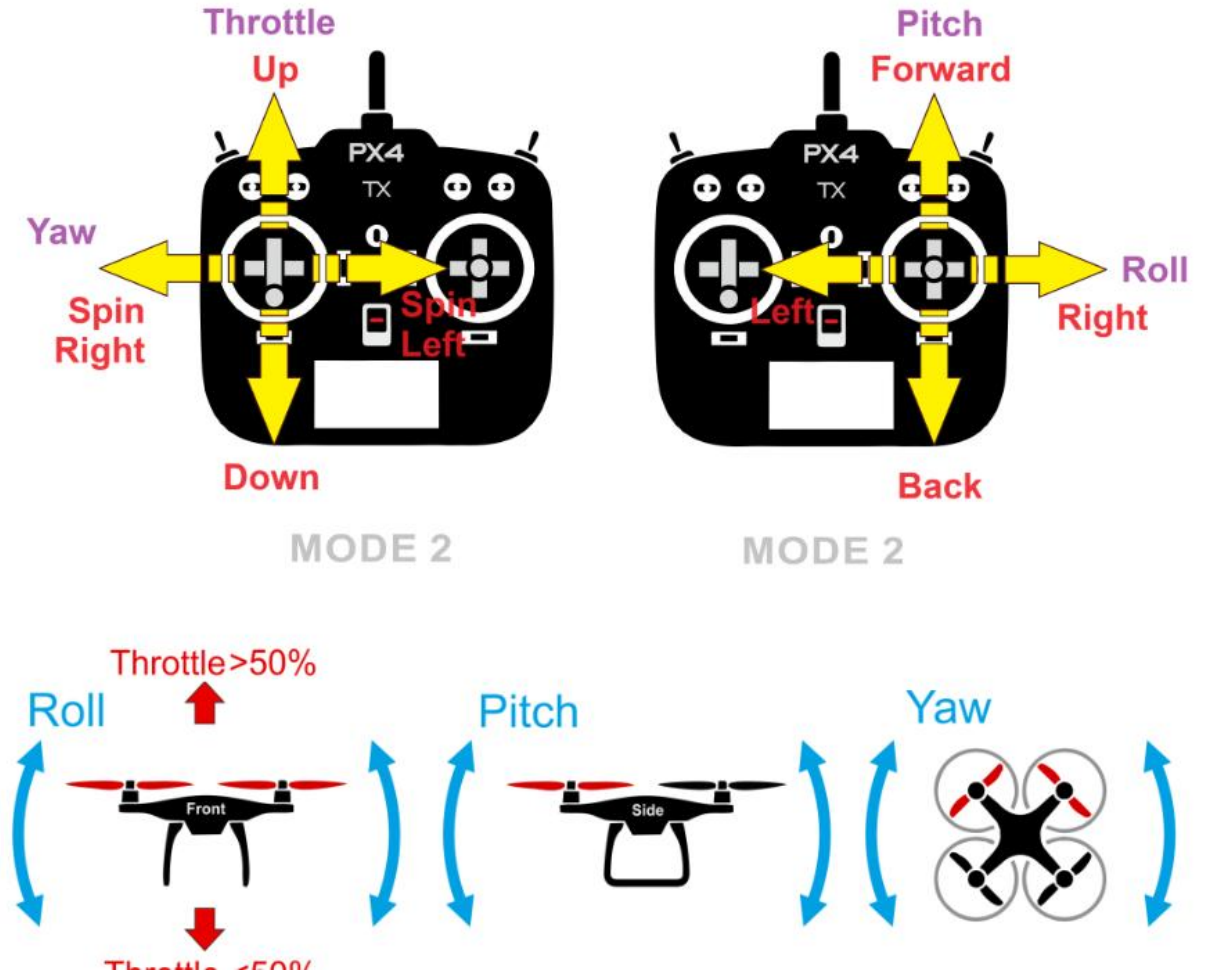
- Flight Modes können manuell oder autonom sein
- Manual
 - Position mode
 - Easiest and safest manual mode for vehicles that have a position fix/GPS.
 - The roll and pitch sticks control acceleration over ground in the vehicle's forward-back and left-right directions, the yaw stick controls horizontal rotation, and the throttle controls speed of ascent-descent.
 - Releasing sticks levels the vehicle, actively brakes it to a stop, and locks it to the current 3D position (even against wind and other forces).
 - Altitude mode
 - Stabilized mode
 - Acro



https://docs.px4.io/v1.16/en/flying/basic_flying_mc

Modes & Arming

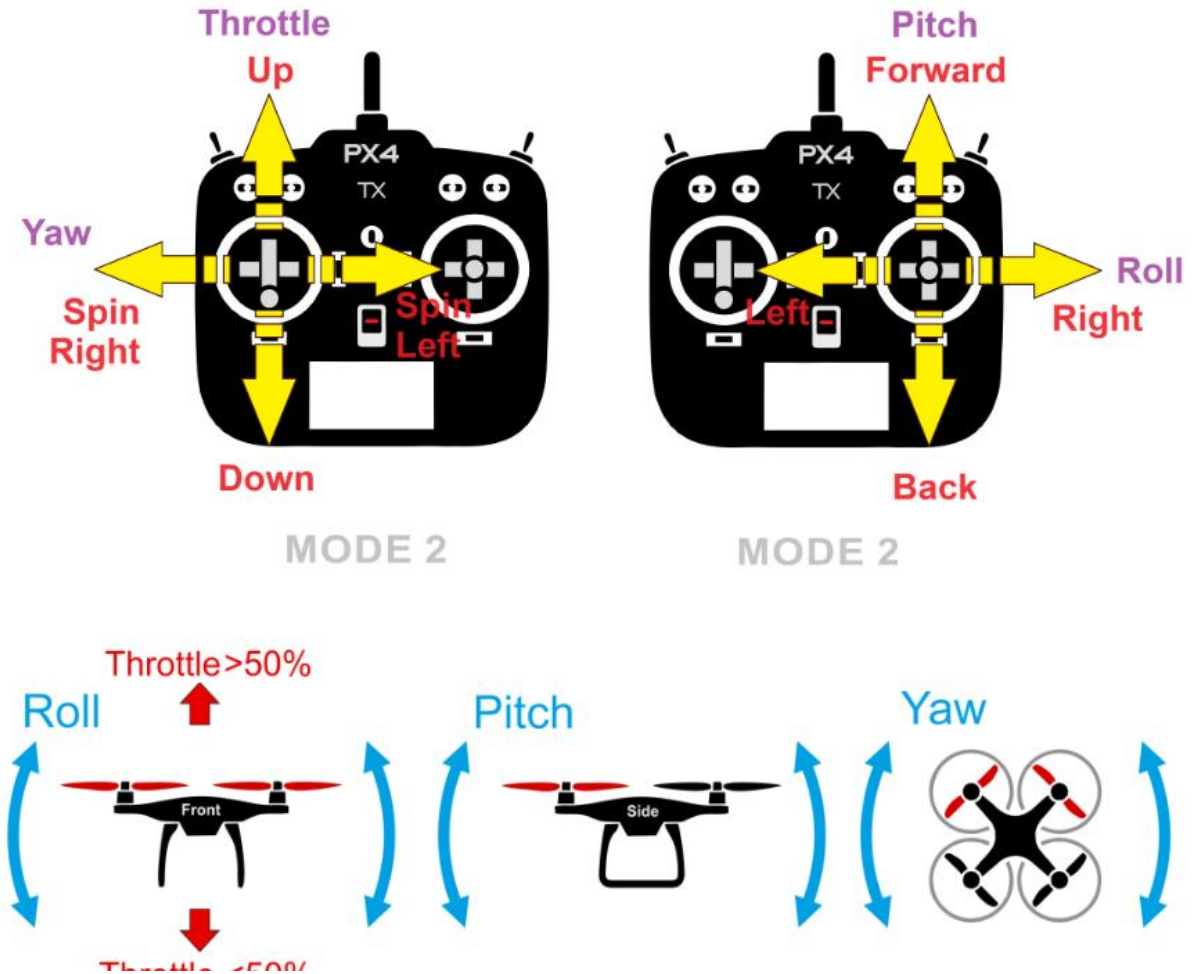
- Flight Modes können manuell oder autonom sein
- Manual
 - Position mode
 - Altitude mode
 - Easiest and safest non-GPS manual mode. The main difference when compared to Position mode is that when the sticks are released the vehicle will level and maintain altitude, but there is no active breaking or holding of horizontal position (the vehicle moves with it's current momentum and drifts with wind).
 - Stabilized mode
 - Acro



https://docs.px4.io/v1.16/en/flying/basic_flying_mc

Modes & Arming

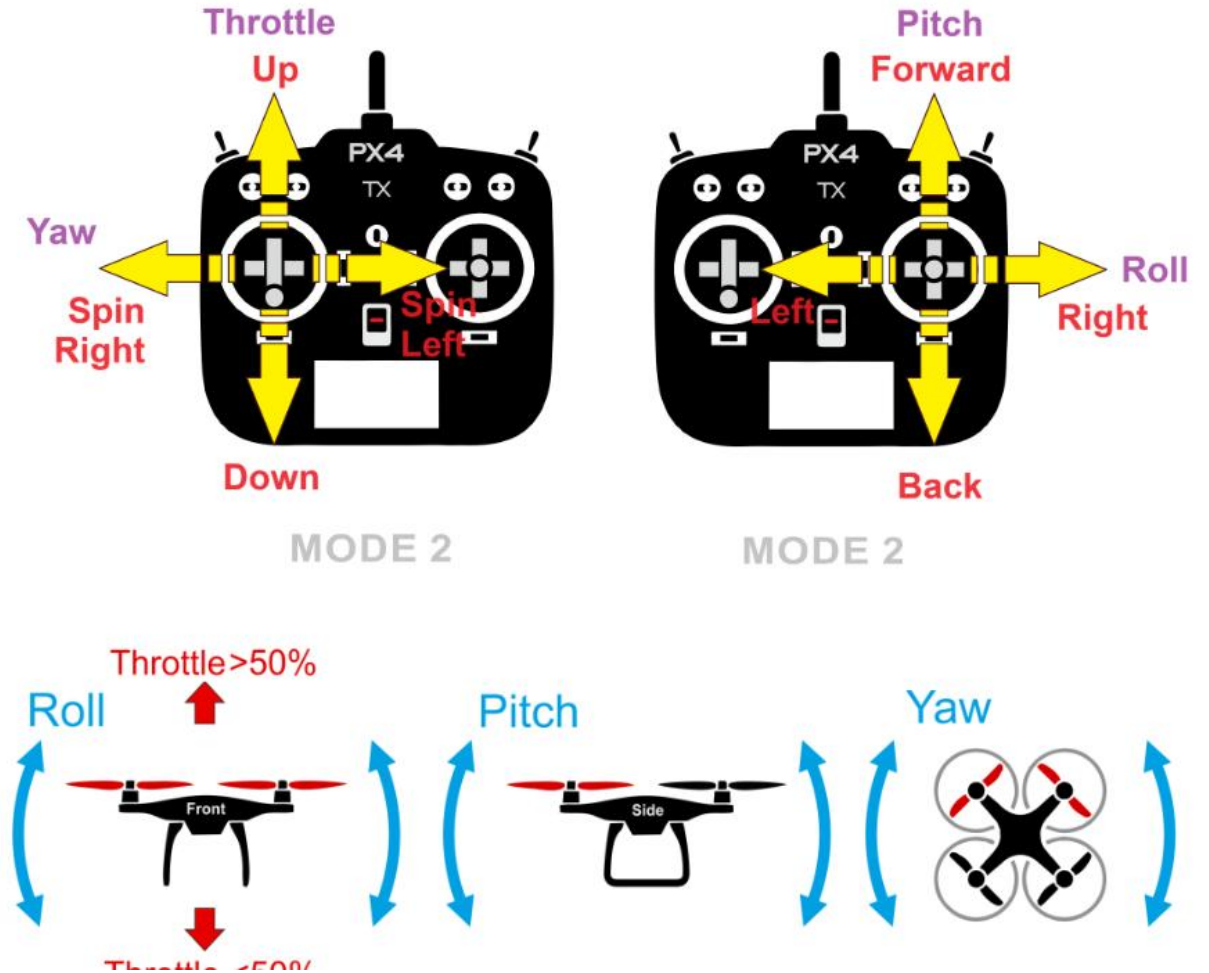
- Flight Modes können manuell oder autonom sein
- Manual
 - Position mode
 - Altitude mode
- Stabilized mode
 - Releasing the sticks levels and maintains the vehicle horizontal posture (but not altitude or position).
 - The vehicle will continue to move with momentum, and both altitude and horizontal position may be affected by wind.
 - This mode is also used if "Manual mode" is selected in a ground station.
- Acro



https://docs.px4.io/v1.16/en/flying/basic_flying_mc

Modes & Arming

- Flight Modes können manuell oder autonom sein
- Manual
 - Position mode
 - Altitude mode
 - Stabilized mode
- Acro
 - Manual mode for performing acrobatic maneuvers, such as rolls and loops. Releasing the sticks **stops the vehicle rotating** in the roll, pitch, yaw axes, but does not otherwise stabilize the vehicle.



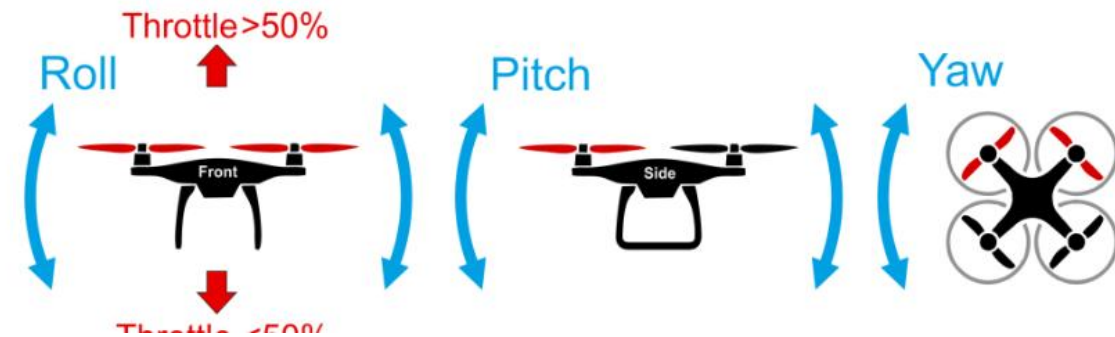
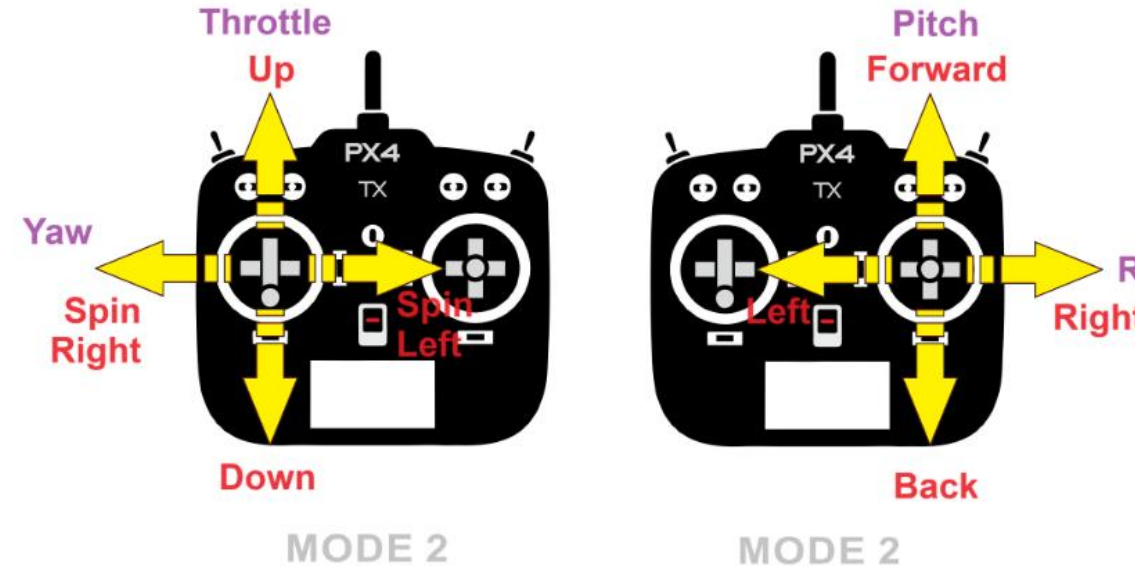
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Modes & Arming

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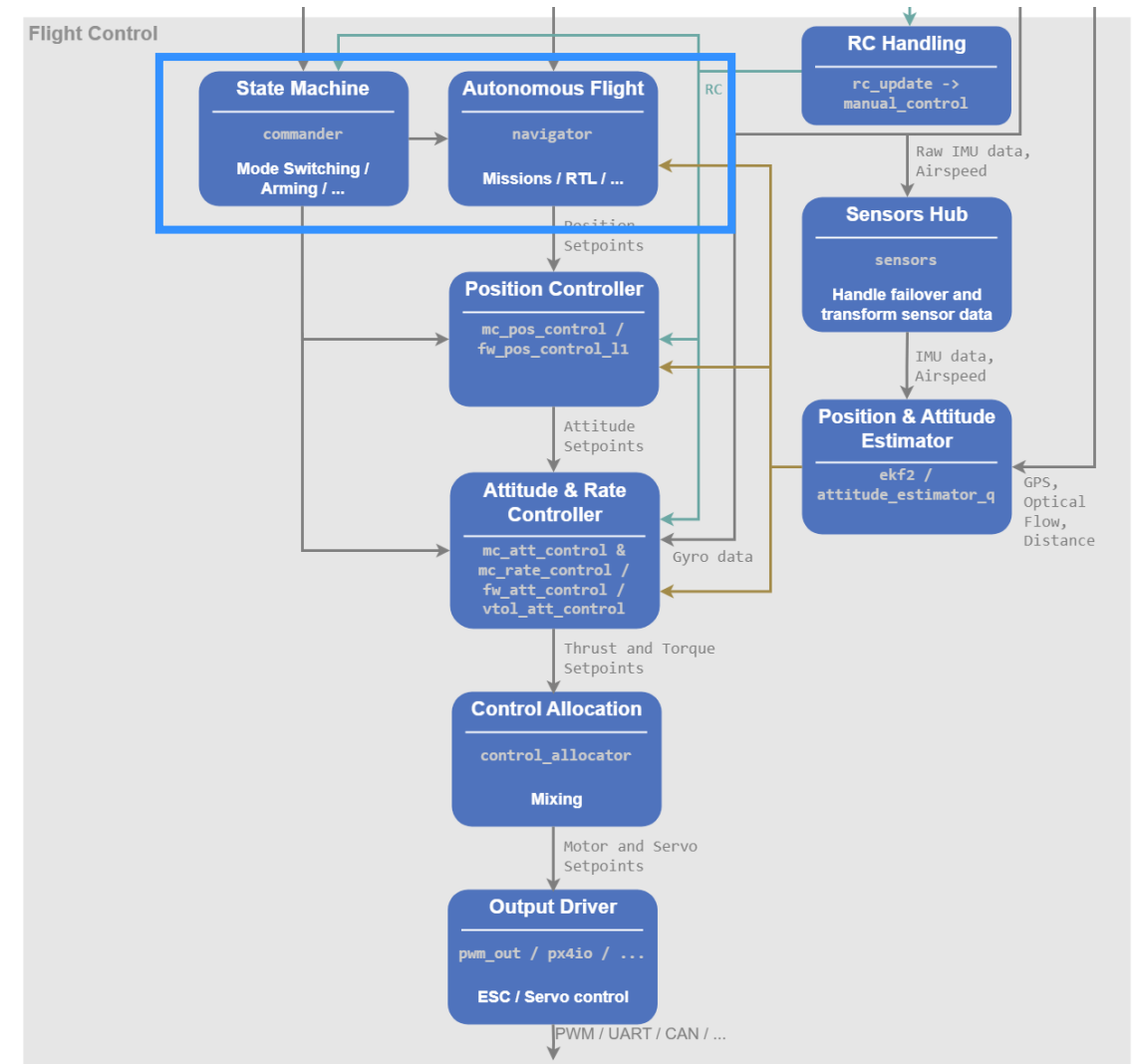
<https://tinyurl.com/SummerCampTHI2>

<https://claude.ai/code/artifact/ec9db891-ceb3-4701-b583-18cda5647966?sk=mEAXbnptA5IJ9XFvI4FqVA>



Regelungsarchitektur: Flight Stack

- Flight Modes können manuell oder autonom sein
- Manual
- Autonom
 - Hold
 - Return:
 - Vehicle ascends to a safe altitude, flies a clear path to a safe location (home or a rally point) and then lands. This requires a global position estimate (GPS).
 - Mission:
 - Vehicle executes a predefined mission/flight plan that has been uploaded to the flight controller. This requires a global position estimate (GPS).
 - Takeoff
 - Land
 - Vehicle lands immediately.
 - Orbit
 - Follow Me
 - Offboard:
 - Vehicle obeys position, velocity, or attitude, setpoints provided via MAVLink or ROS 2.

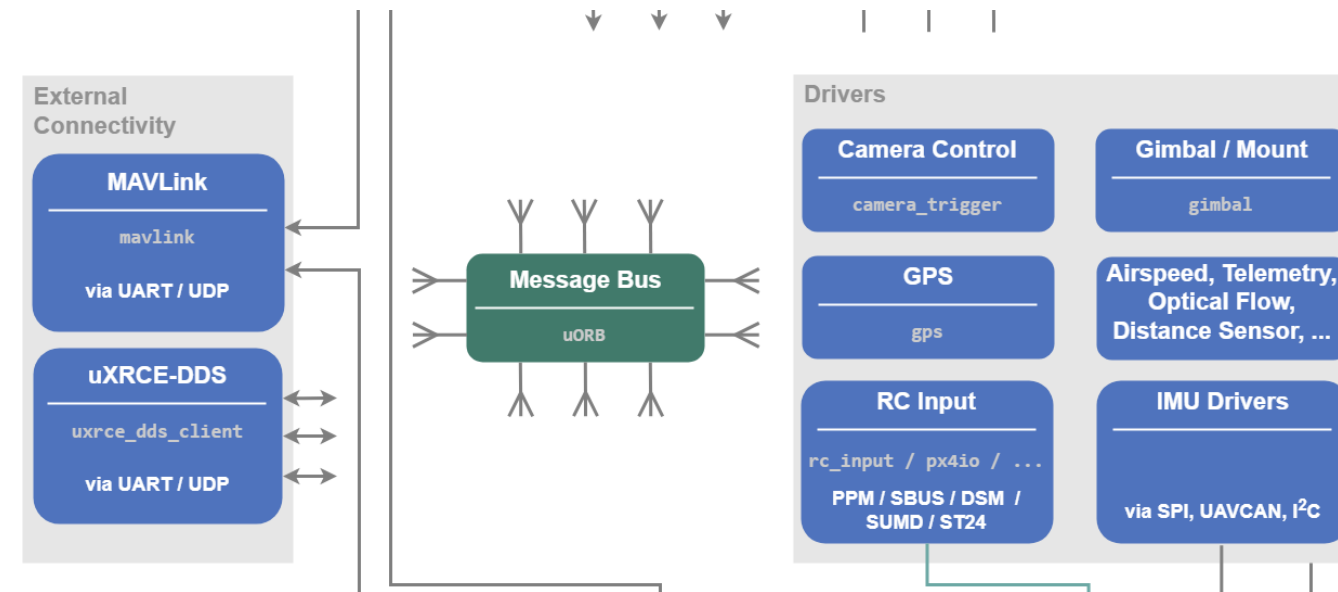


<https://docs.px4.io/main/en/concept/architecture>

Systemarchitektur: Middleware

Systemarchitektur am Beispiel von Px4

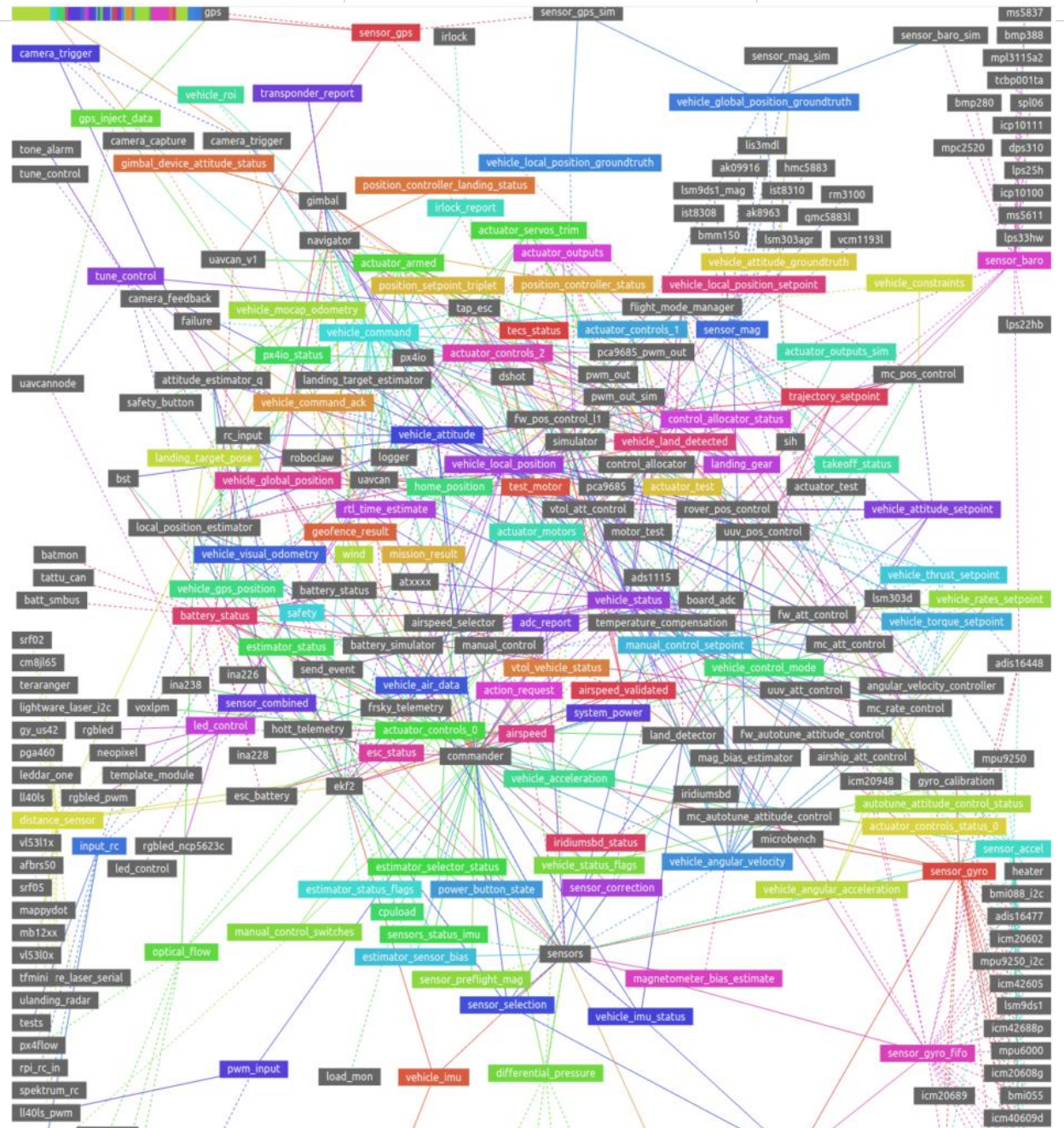
- Flight Control
- **Middleware**
 - HW-Driver
 - **Message Bus**
 - Simulation Layer
 - Storage
 - External Connectivity



<https://docs.px4.io/main/en/concept/architecture>

Systemarchitektur: Message Bus

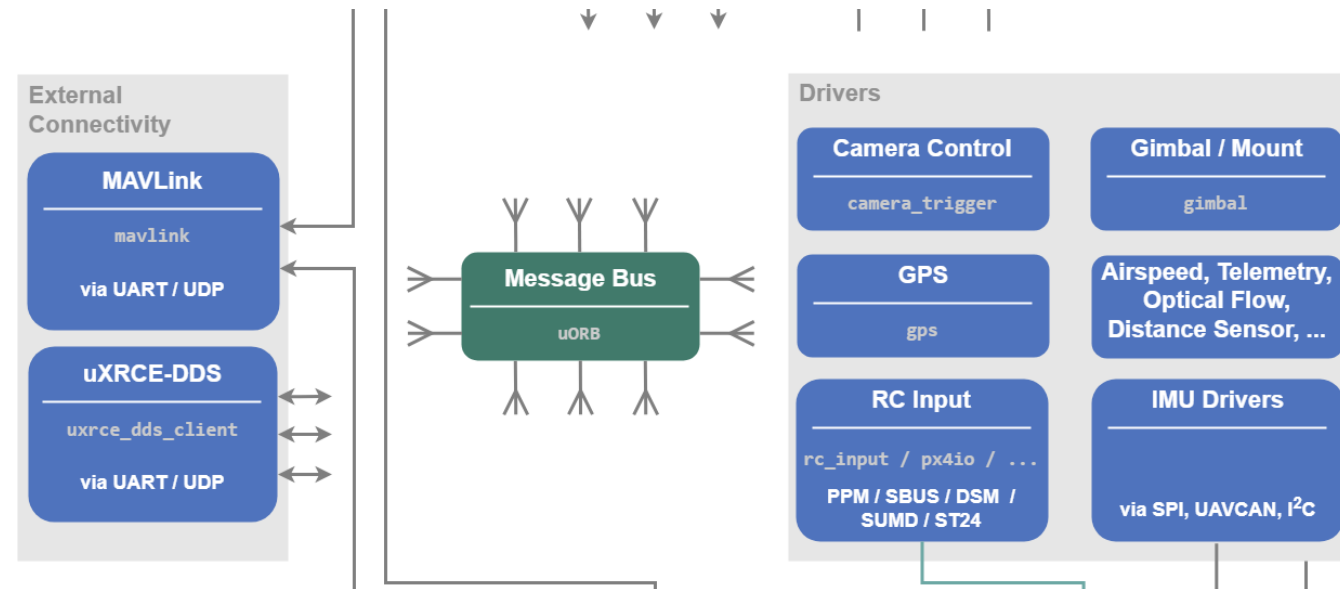
- Current uORB messaging topology of PX4 in a single view.
- Wer redet mit wem?



<https://px4.io/px4-uorb-explained-part-1/>

Systemarchitektur: Message Bus

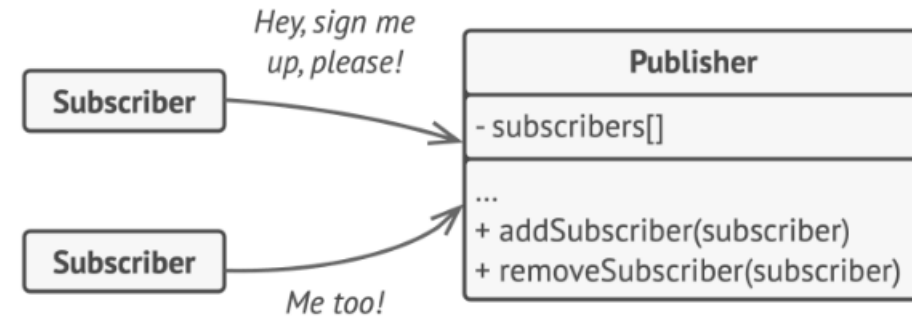
- Typisches Problem in Software:
 - Wie bekomme ich bei komplexer Software oder Multi-Thread-Anwendungen Daten von einem zum anderen Prozess?
 - Wie kann ich auf neue Sensordaten reagieren?
- Lösungsvorschläge:
 - Singleton-Pattern (Zentraler Datencontainer)
 - Mediator Pattern
 - Observer- / Publish-Subscribe-Pattern
 - ...



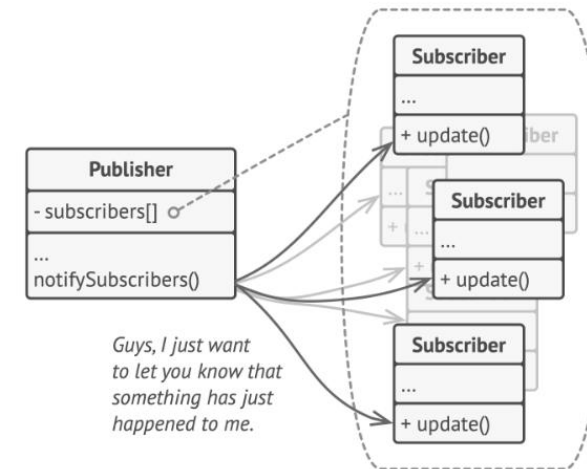
<https://docs.px4.io/main/en/concept/architecture>

Message Bus: Observer- / Publish-Subscribe-Pattern

- Im Code können bestimmte Nachrichten abonniert werden
- Nachrichten können (an anderer Stelle) in zentrale Instanz *published* werden
- Ein *published* Update verursacht die Benachrichtigung der Abonnenten
- Das Messagingsystem von ROS2 funktioniert nach einem ähnlichen Pattern



A subscription mechanism lets individual objects subscribe to event notifications.



Publisher notifies subscribers by calling the specific notification method on their objects.

<https://refactoring.guru/design-patterns/observer>

Systemarchitektur: Message Bus

- *In PX4:*

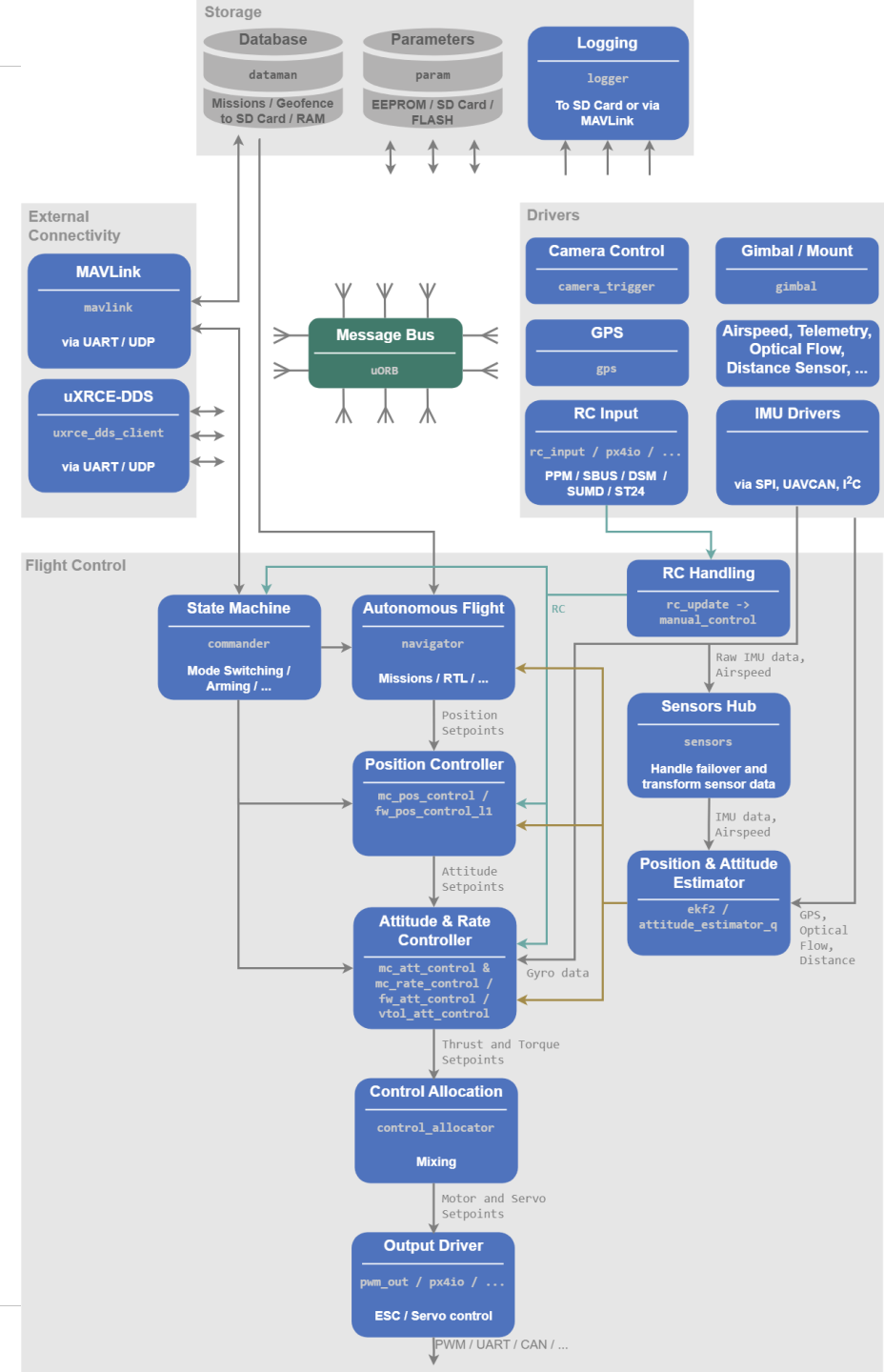
- https://github.com/PX4/PX4-Autopilot/blob/main/src/examples/work_item/WorkItemExample.cpp

Systemarchitektur

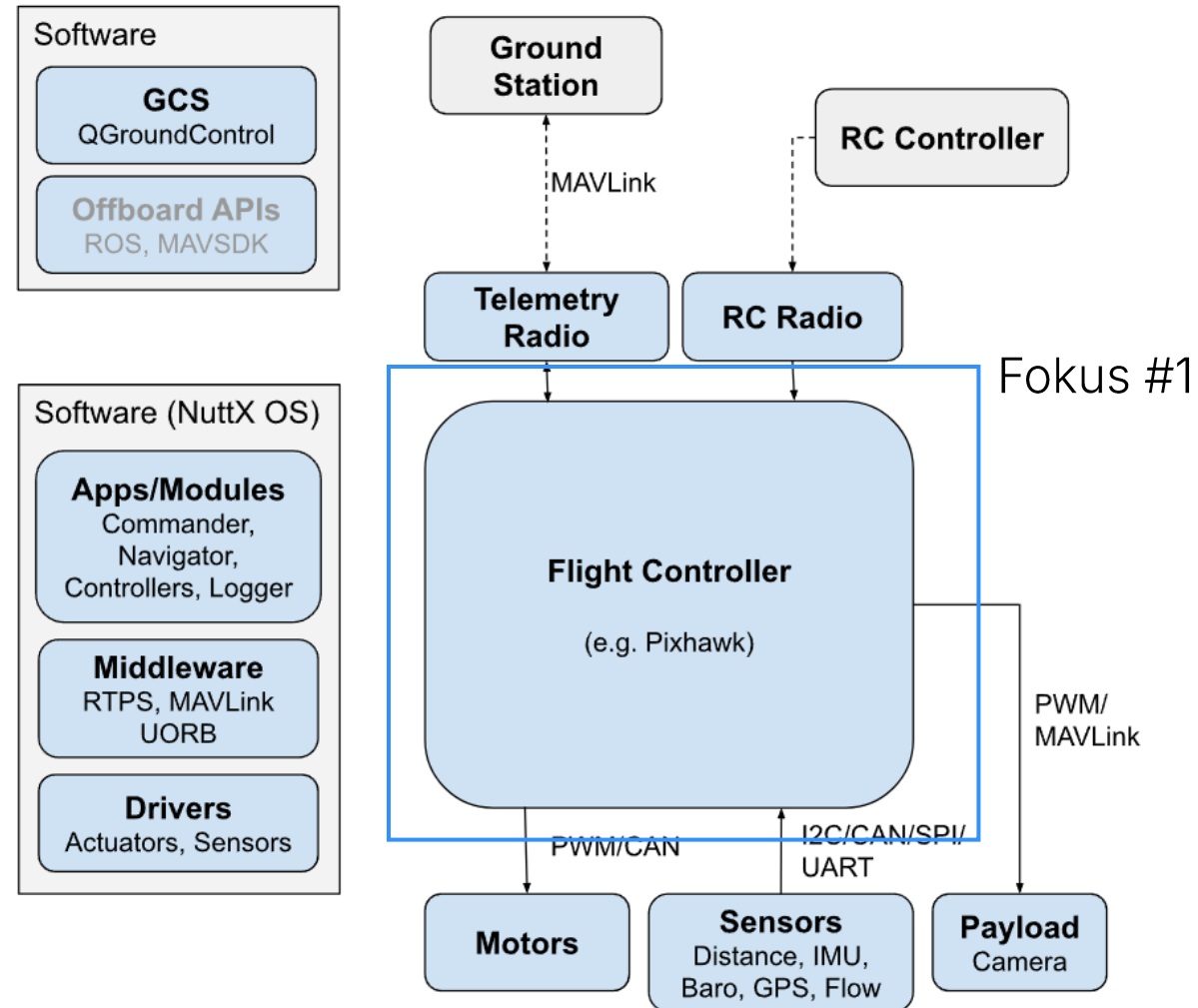
Systemarchitektur am Beispiel von Px4

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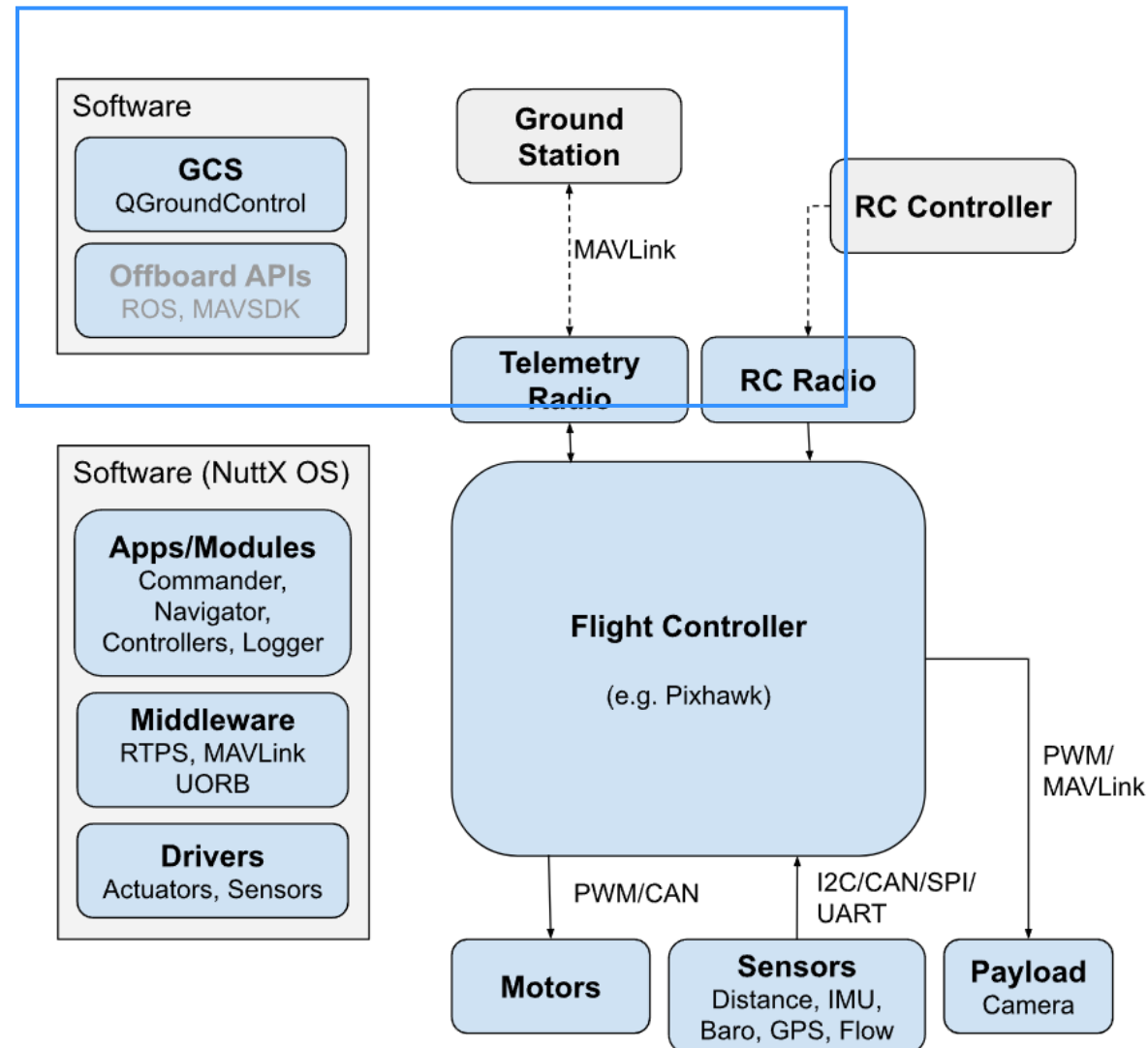


System Architecture



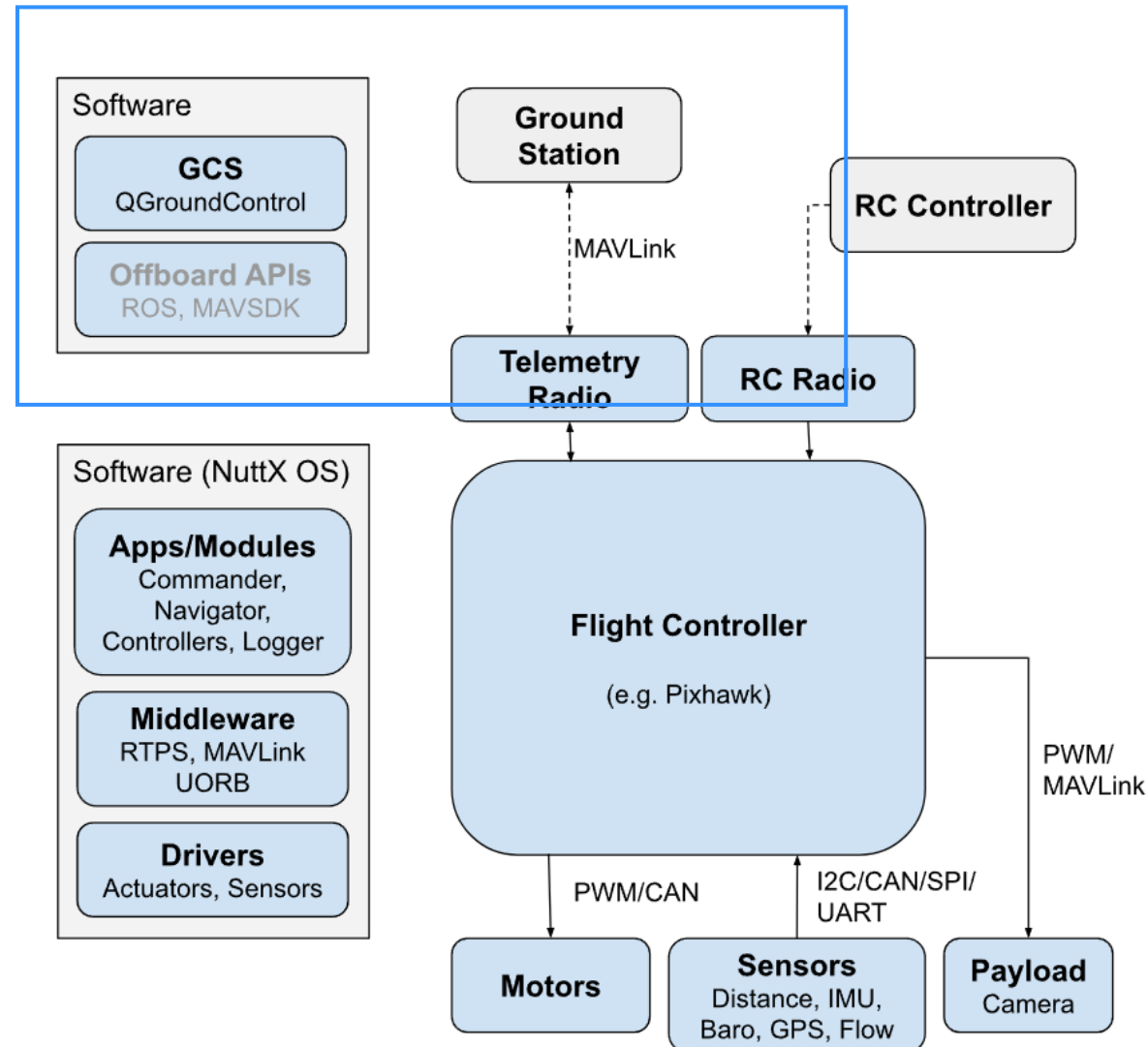
System Architecture: Standard Setup (FC only)

Fokus #2



System Architecture: Standard Setup (FC only)

Fokus #2



Standardkommunikation Ground ↔ Drone: MAV-Link Protocol

- Leichtgewichtiges Nachrichtenprotokoll für die Kommunikation mit Drohnen und zwischen deren Onboard-Komponenten
- Datenströme werden als Topics publiziert, Konfigurations-Subprotokolle wie Mission oder Parameter laufen Punkt-zu-Punkt mit Wiederholendung
- Sehr geringer Overhead – 8 Byte pro Paket bei MAVLink 1, 12 Byte bei MAVLink 2, das dafür sicherer und erweiterbar ist

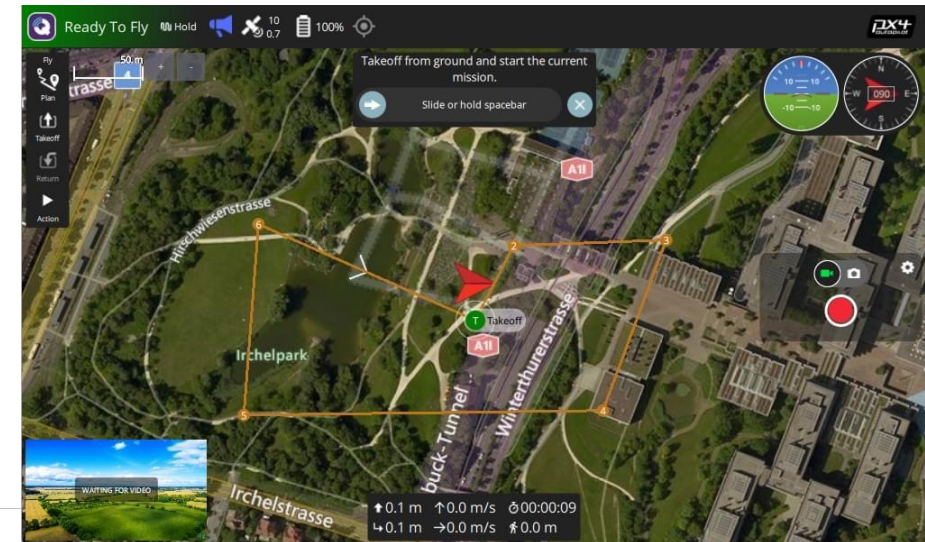
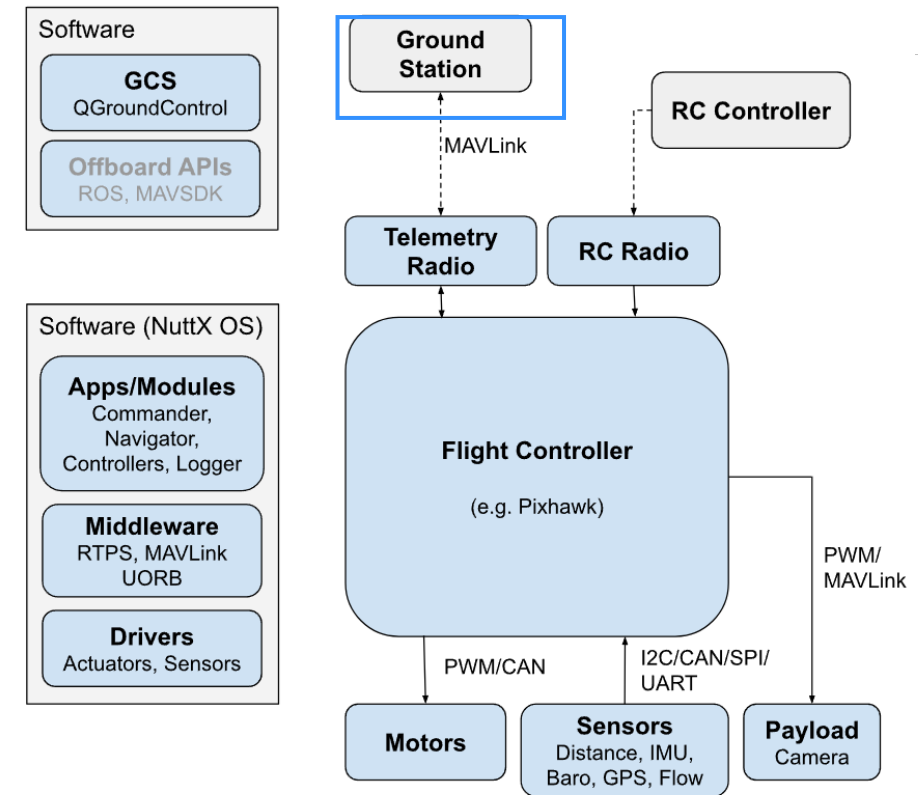
<https://mavlink.io/en/messages/common.html>



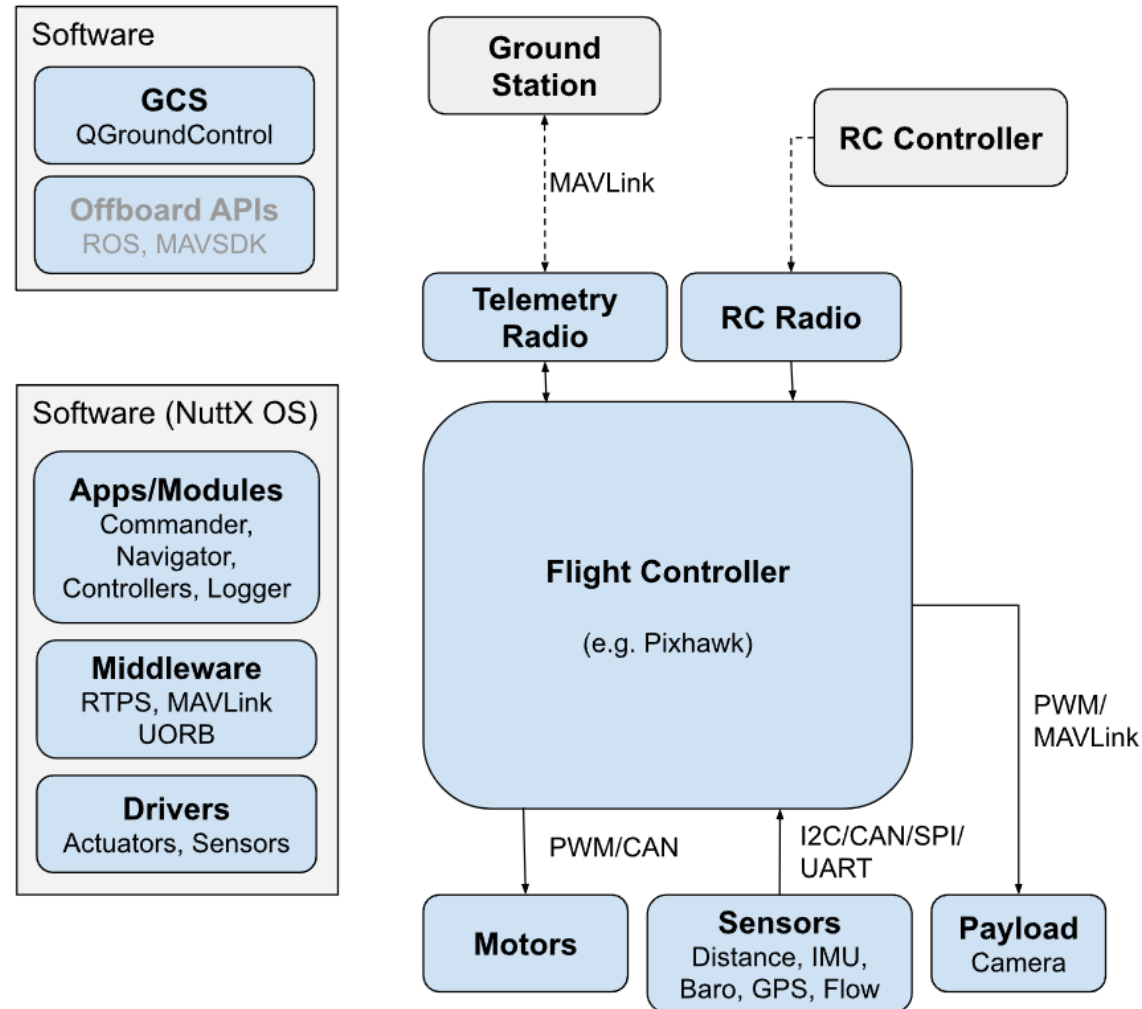
<https://mavlink.io/en/>

QGroundControl/MissionPlanner

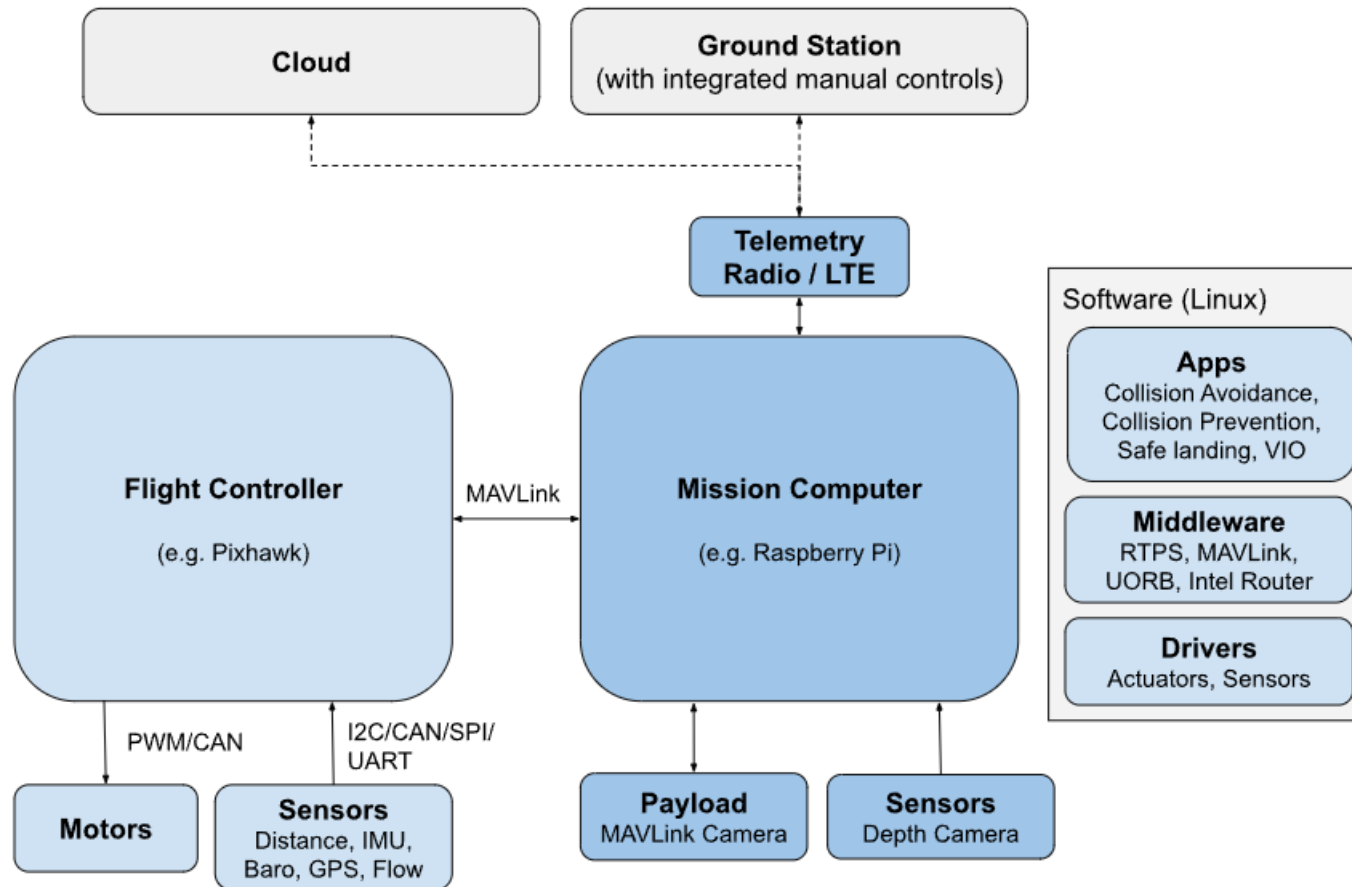
- Software für
 - Konfiguration von Flugreglern (daher meist an zugrundeliegenden FC gebunden)
 - Vorplanung von Missionen
 - Überwachung von Missionen
 - Logging/Auswertung



System Architecture: Standard Setup (FC only)



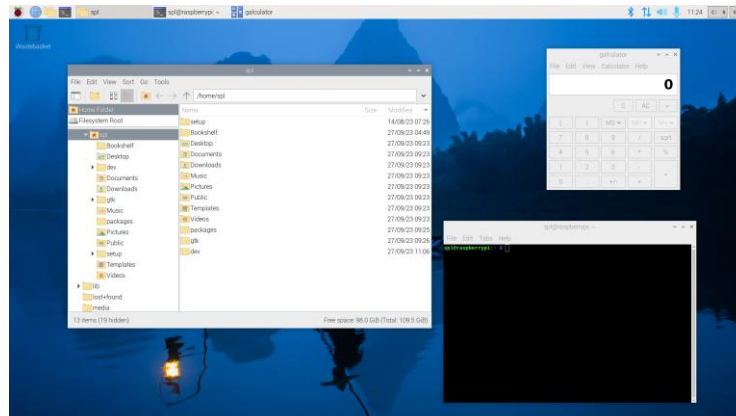
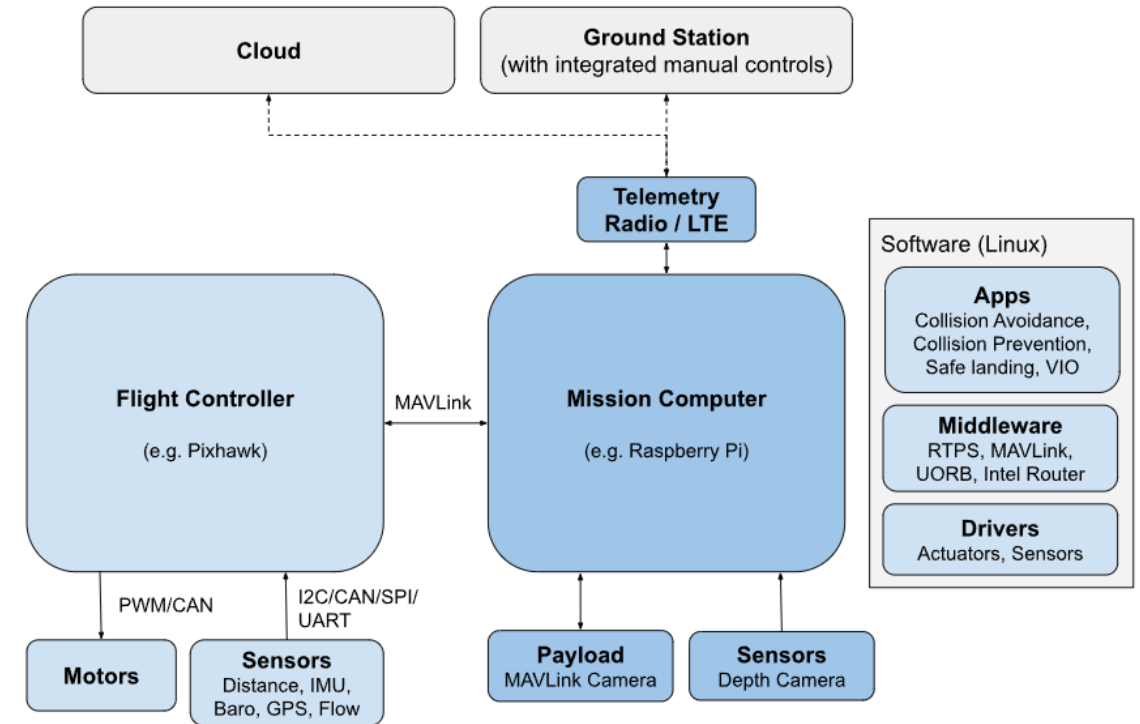
System Architecture: Extended Setup (Companion Computer)



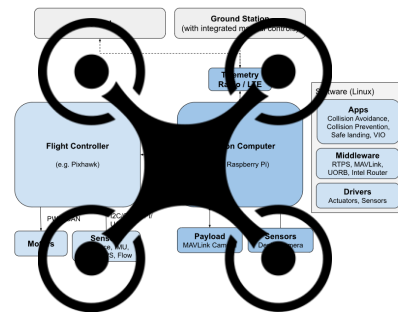
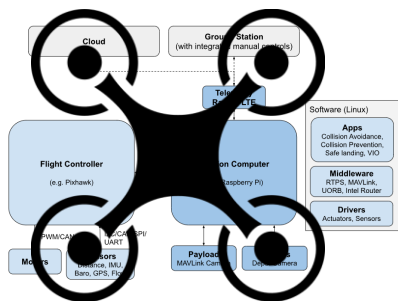
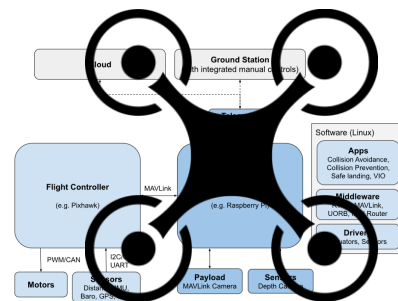
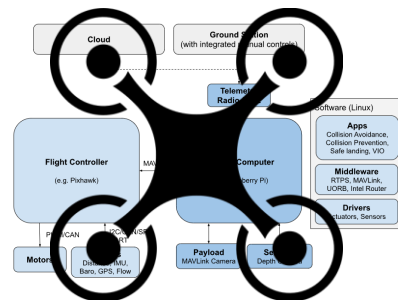
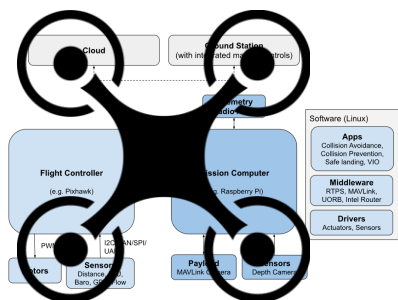
System Architecture: Extended Setup (Companion Computer)

- MAV-Link Verbindung wird über Missionscomputer in den Flight Controller gesendet
- Missionscomputer bietet neue Möglichkeiten
 - der Automatisierung
 - der Sensordatenverarbeitung
 - der Konnektivität
 - ...

- Bspw. Raspberry



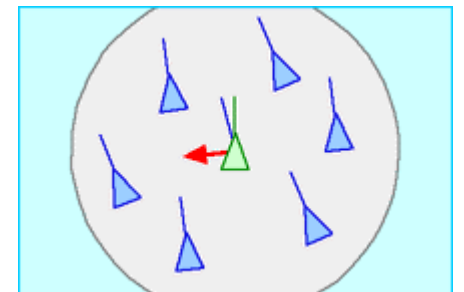
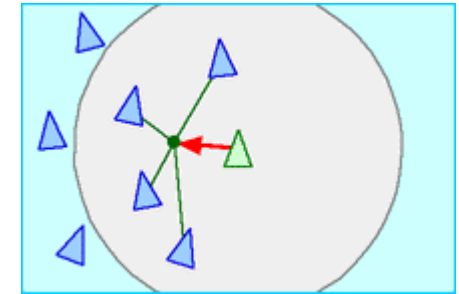
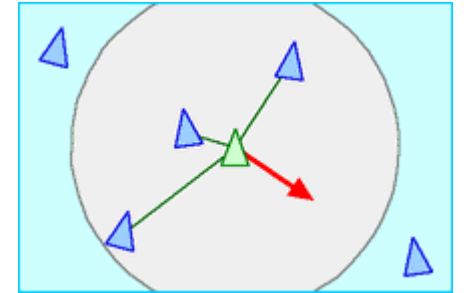
Multi-UAV Applications



Welche Fragen der Systemarchitektur ergeben sich aus Multi-UAV Anwendungen?

Classical Swarming Algorithms: Boids

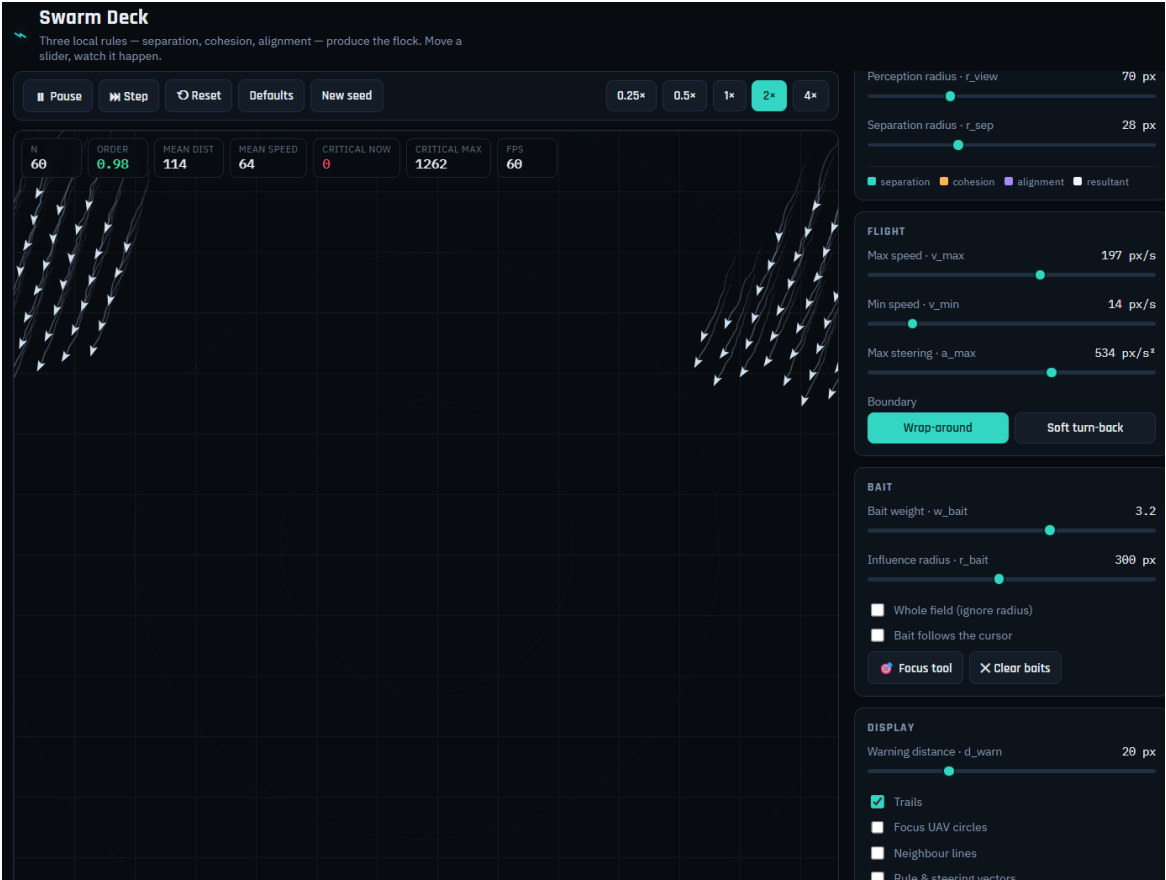
- Craig Reynolds (1987): Schwarmverhalten als emergentes Verhalten ohne zentrale Steuerung, globalen Plan oder Führer.
- Jedes „Boid“ sieht nur Nachbarn innerhalb eines Radius (+ Sichtfeldwinkel)
- Drei Regeln werden summiert um Beschleunigung in jedem Zeitschritt zu berechnen
 - Separation
 - Cohesion.
 - Alignment
- Schwarmverhalten resultiert aus der Gewichtung der drei Faktoren



Boid Simulation

<https://claude.ai/artifact/REhaEMCbJHFspueFQrCWyC>

<https://tinyurl.com/SummerCampTHI3>



Inspired by: <https://eater.net/boids>

Recap: UxS-Schwärme

Einige Vorschläge zur Definition:

- „An assembly that operates cooperatively to commands from a single operator in charge through a common architecture. The three load-bearing elements are size ("assembly"), interaction ("operates cooperatively"), and span of control ("single operator in charge") [1]“
- „Ein „Schwarm“ bezeichnet umgangssprachlich einen großen, einheitlich formierten dreidimensionalen meist anonymen Verband flugfähiger oder wasserbewohnender Tiere wie Vögel, Fische, Insekten oder Krebse, deren Mitglieder sich durch das Einhalten bestimmter Positionen und eine angeborene soziale Appetenz zum Gesamtverband räumlich und zeitlich eng koordiniert verhalten (Lexikon der Biologie)“
- „A network of several homogeneous platforms that pursue a dedicated task by means of decentralised, temporally and spatially limited coordination. The execution of the order only comes about through the interaction. [2]“